

Exercise and worked solution

Linear regression

Four parts to master OLS regression end to end: estimating the parameters, inference on the slope, diagnosing a model (omitted variable, multicollinearity), and reading heteroskedasticity. Work through each part before reading the solution.

Part A · Estimating an OLS line

PROBLEM

We regress a sector's return (Y) on the market's (X) over 20 years: We measure a covariance $\sigma_{XY} = 512.3$, a variance $\sigma_X^2 = 337.3$, a mean $\bar{y} = 15.4\%$ and $\bar{x} = 7.77\%$. (a) Compute the slope β . (b) Compute the intercept α and interpret it.

SOLUTION

- (a) $\beta = \sigma_{XY} / \sigma_X^2 = 512.3 / 337.3 = \mathbf{1.52}$. When the market gains 1%, the sector gains on average 1.52%. This is a $\beta > 1$, so a sector more volatile than the market.
- (b) $\alpha = \bar{y} - \beta \bar{x} = 15.4 - 1.52 \times 7.77 = 15.4 - 11.81 = \mathbf{3.60\%}$. This is the sector's mean return when the market is at zero (the CAPM 'abnormal' return). Recall: $\beta = \rho \cdot \sigma_Y / \sigma_X$, so $\beta = 0$ if and only if the correlation is zero.

Part B · Inference on the slope

PROBLEM

The same regression gives $\beta = 1.52$ with a standard error $\text{s.e.}(\beta) = 0.40$. (a) Compute the t-statistic for $H_0: \beta = 0$ and conclude at 5%. (b) Give the 95% CI. (c) The R^2 is 0.44: what does it measure?

SOLUTION

- (a) $t = \beta / \text{s.e.}(\beta) = 1.52 / 0.40 = \mathbf{3.80}$. Since $3.80 > 1.96$, we **reject H_0** at 5%: the slope is significantly non-zero.
- (b) $95\% \text{ CI} = \beta \pm 1.96 \times \text{s.e.} = 1.52 \pm 1.96 \times 0.40 = 1.52 \pm 0.784 = \mathbf{[0.74; 2.30]}$. It excludes 0: same conclusion as the test, second reading. The third ($p\text{-value} \approx 0.01\% < 5\%$) confirms again.
- (c) $R^2 = 0.44$ is the share of Y 's variance explained by X . In simple regression, $R^2 = \rho^2$: here the correlation is $\sqrt{0.44} \approx 0.66$. R^2 is symmetric (regressing Y on X or X on Y gives the same), unlike the slope.

Part C · Omitted variable and multicollinearity

PROBLEM

(a) The true model is $Y = 2 + 1.2 X_1 - 2.1 X_2 + \varepsilon$, with $\delta = \text{Cov}(X_1, X_2) / V(X_1) = 0.60$. We omit X_2 . What value does the estimated coefficient of X_1 converge to? (b) In another model, the correlation between two regressors is $\rho = 0.96$. Compute the VIF and conclude.

SOLUTION

- (a) Omitted variable bias: $\beta_1 \rightarrow \beta_1 + \beta_2 \delta = 1.2 + (-2.1) \times 0.60 = 1.2 - 1.26 = \mathbf{-0.06}$. The bias is so strong the **sign flips**: the true effect (+1.2) appears negative. Two conditions held: X_2 matters ($\beta_2 \neq 0$) and X_1, X_2 are correlated ($\delta \neq 0$).
- (b) With two variables, $R_j^2 = \rho^2 = 0.96^2 = 0.9216$. $\text{VIF} = 1 / (1 - R_j^2) = 1 / 0.0784 \approx \mathbf{12.8}$. Since $12.8 > 10$, multicollinearity is excessive. It is not a technical fault (no assumption violated), just a readability problem: the F can stay significant while individual t's are weak.

Part D · Heteroskedasticity and robustness

PROBLEM

An analyst tests a portfolio's *size* factor. Under homoskedasticity, the t is -3.2 (significant). With robust standard errors, the t drops to -1.3 . (a) What changed, and why? (b) Are the coefficients biased? (c) Which conclusion should be kept?

SOLUTION

1. (a) The data are **heteroskedastic** (shock variance depending on X , detectable by the White test). The usual s.e. are then too small, artificially inflating the t -stat; robust s.e. (larger) correct this and the t -stat falls.
2. (b) **No**. Heteroskedasticity leaves the coefficients unbiased and consistent: only the standard errors are wrong. This is the classic trap — answering 'biased estimators' is incorrect.
3. (c) With -1.3 , below 1.96 in absolute value, the factor is **no longer significant**. Keep the robust conclusion. If the five assumptions hold, OLS is BLUE; adding error normality, it becomes BUE.

Key takeaways: $\beta = \text{cov}/\text{var} = \rho \cdot \sigma_y / \sigma_x$ (so $\beta = 0 \iff \rho = 0$); test with $t = \beta/\text{s.e.}$, with the equivalence $|t| > 1.96 \iff p < 5\% \iff 0 \notin \text{CI}$; omitting a correlated variable biases the coefficients; $\text{VIF} > 10$ signals multicollinearity; and heteroskedasticity distorts the s.e. but never the coefficients.